

MINISTRY OF EDUCATION AND TRAINING
EASTERN INTERNATIONAL UNIVERSITY



MIS 395
ARTIFICIAL INTELLIGENCE FOR BUSINESS

Final Project

**Multidomain Analytics for Strategic
Decision-Making**

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I. Project Motivation

As artificial intelligence (AI) continues to transform the business landscape, its integration into strategic decision-making processes is becoming increasingly essential. This project is motivated by the need to harness AI-powered tools across diverse business domains - namely marketing, retail operations, and human resources - to uncover actionable insights and drive performance improvements.

In the marketing domain, we utilize the Bank Marketing dataset from the UCI Machine Learning Repository, which provides detailed records of direct marketing campaigns. The application of AI to this data enables organizations to predict customer responses, enhance targeting strategies, and improve campaign effectiveness.

In the realm of retail analytics, the Bike Sales dataset from Kaggle offers over 100,000 transaction records. AI techniques applied to this data allow businesses to detect sales patterns, forecast demand, and optimize inventory planning - critical factors in maintaining operational efficiency and profitability.

In human resource management, the Employee Performance and Productivity dataset provides insights into workforce behavior and outcomes. Through AI-driven analysis, we aim to identify performance trends, evaluate productivity drivers, and support evidence-based talent management strategies.

By combining these three datasets, this project highlights the cross-functional value of AI and its potential to streamline workflows, personalize services, and reduce operational costs. Each dataset represents a unique business scenario where AI can assist in making faster, data-informed decisions. The project ultimately seeks to demonstrate that AI is not just a technological trend but a fundamental enabler of modern business intelligence.

II. Dataset Overview

No	Dataset Name	Domain	No. of Records	No. of Features	Model Applied	Missing Value?
3	Employee Performance & Productivity	Human Resources	100,000	20	Classification	No

Project objective: Build a classification model that can predict which employees are likely to resign. This allows the company to move from a reactive to a proactive approach, allowing for timely interventions to retain talent and save costs.

III. Tools Used and Why

For this project, a multi-tool strategy was adopted to leverage the specific strengths of different platforms for various stages of the data analysis workflow. The selection of each tool was based on its suitability for the task at hand, from initial exploration to final presentation.

Python (Google Colab)

Python, with its extensive ecosystem of libraries such as Pandas, Matplotlib, Seaborn, and Scikit-learn, served as the primary tool for the project's in-depth technical execution. It was utilized for all detailed data preprocessing tasks, including feature scaling and engineering; for generating custom visualizations for exploratory data analysis; and for building, training, and rigorously evaluating the two binary classification models. The choice of Python was driven by the need for precise control over the analytical process and access to powerful, industry-standard machine learning libraries.

Julius AI

Julius AI was employed at the beginning of the project as a rapid exploratory data analysis tool. Its natural language interface was used to generate initial descriptive statistics and visualizations from the dataset. This approach was chosen to quickly form a baseline

understanding of the data's structure and to generate initial hypotheses without the immediate need for coding.

Graphite Note

Graphite Note was utilized as a no-code predictive analytics platform to independently build a binary classification model and to present the project's final results. This tool was selected to demonstrate a modern, business-focused approach to AI, where users can leverage predictive analytics without deep coding expertise. Its automated features for preprocessing and modeling were used to generate a predictive outcome, and its data storytelling capabilities were employed to structure and communicate the project's key findings in a clear, narrative-driven format.

IV. Key Findings (with Visuals)

1. Descriptive analysis with Python

df.describe()

	Employee_ID	Age	Years_At_Company	Performance_Score	Monthly_Salary	Work_Hours_Per_Week	Projects_Handled	Overtime_Hours	Sick_Days	Remote_Work_Frequency	Team_Size	Training_Hours	Promotions	Employee_Satisfaction_Score
count	100000.000000	100000.000000	100000.000000	100000.000000	100000.000000	100000.000000	100000.000000	100000.000000	100000.000000	100000.000000	100000.000000	100000.000000	100000.000000	100000.000000
mean	50000.500000	41.029410	4.476070	2.995430	6403.211000	44.956950	24.431170	14.514930	7.008550	50.090500	10.013560	49.506060	0.999720	2.999088
std	28867.657797	11.244121	2.869336	1.414726	1372.508717	8.942003	14.469584	8.664026	4.331591	35.351157	5.495405	28.890383	0.815872	1.150719
min	1.000000	22.000000	0.000000	1.000000	3850.000000	30.000000	0.000000	0.000000	0.000000	0.000000	1.000000	0.000000	0.000000	1.000000
25%	25000.750000	31.000000	2.000000	2.000000	5250.000000	37.000000	12.000000	7.000000	3.000000	25.000000	5.000000	25.000000	0.000000	2.010000
50%	50000.500000	41.000000	4.000000	3.000000	6500.000000	45.000000	24.000000	15.000000	7.000000	50.000000	10.000000	49.000000	1.000000	3.000000
75%	75000.250000	51.000000	7.000000	4.000000	7500.000000	53.000000	37.000000	22.000000	11.000000	75.000000	15.000000	75.000000	2.000000	3.990000
max	100000.000000	60.000000	10.000000	5.000000	9000.000000	60.000000	49.000000	29.000000	14.000000	100.000000	19.000000	99.000000	2.000000	5.000000

The initial descriptive analysis provided a foundational understanding of the dataset's structure and the characteristics of the employees. Key findings from this stage include:

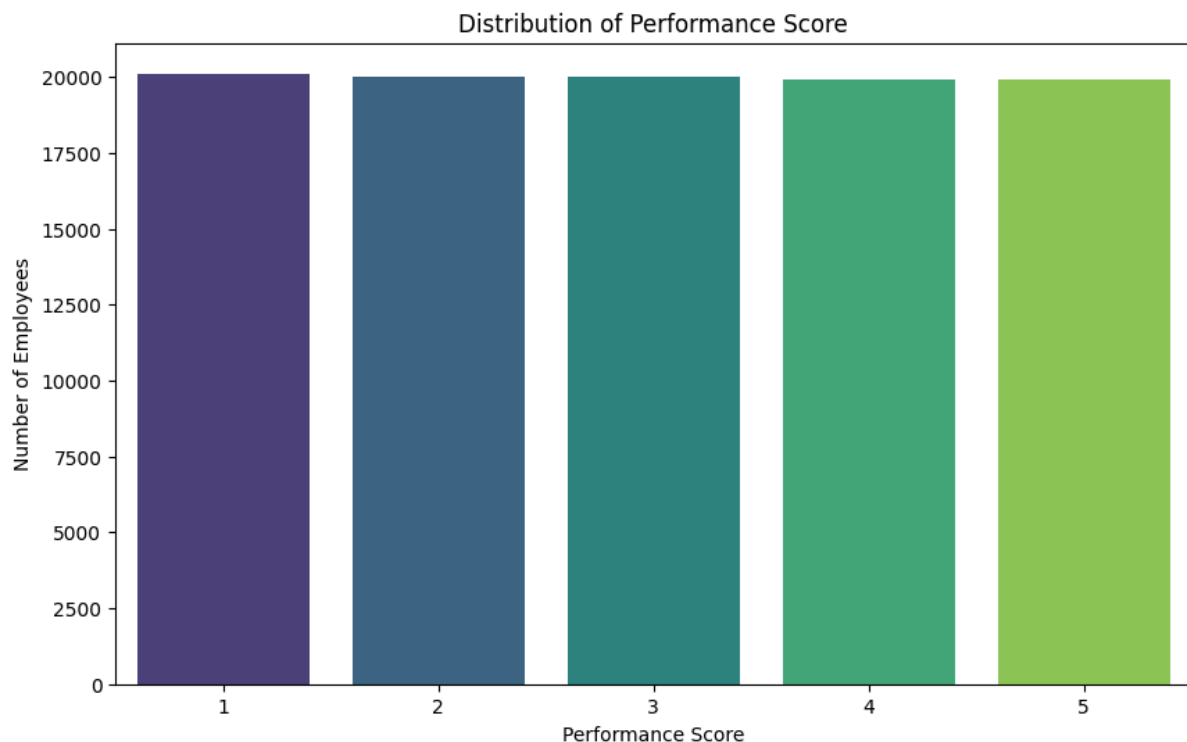
- **Dataset Overview:** The dataset comprises approximately 100,000 employee records, detailing information across various professional and demographic attributes.
- **Target Variable Imbalance:** The target variable, Resigned, exhibits a significant class imbalance, with a small percentage of employees having resigned (Resigned =

True). This highlights the necessity of using specialized techniques and metrics to accurately model this minority class.

- **Demographic and Employment Distribution:** The average employee in the dataset is around **40 years old** and has worked at the company for approximately **5 years**. Employees are distributed across various departments, with **IT, Sales, and Customer Support** being the most prevalent.
- **Performance and Satisfaction:** The distribution of `Performance_Score` is relatively varied, with a notable number of employees scoring in the mid-range. The `Employee_Satisfaction_Score` also shows a wide range, indicating diverse levels of workplace contentment.
- **Workload and Compensation:** The average `Monthly_Salary` and `Work_Hours_Per_Week` provide a baseline for compensation and workload. A significant number of employees report working overtime, which is a key factor to be further investigated for its potential correlation with other metrics.

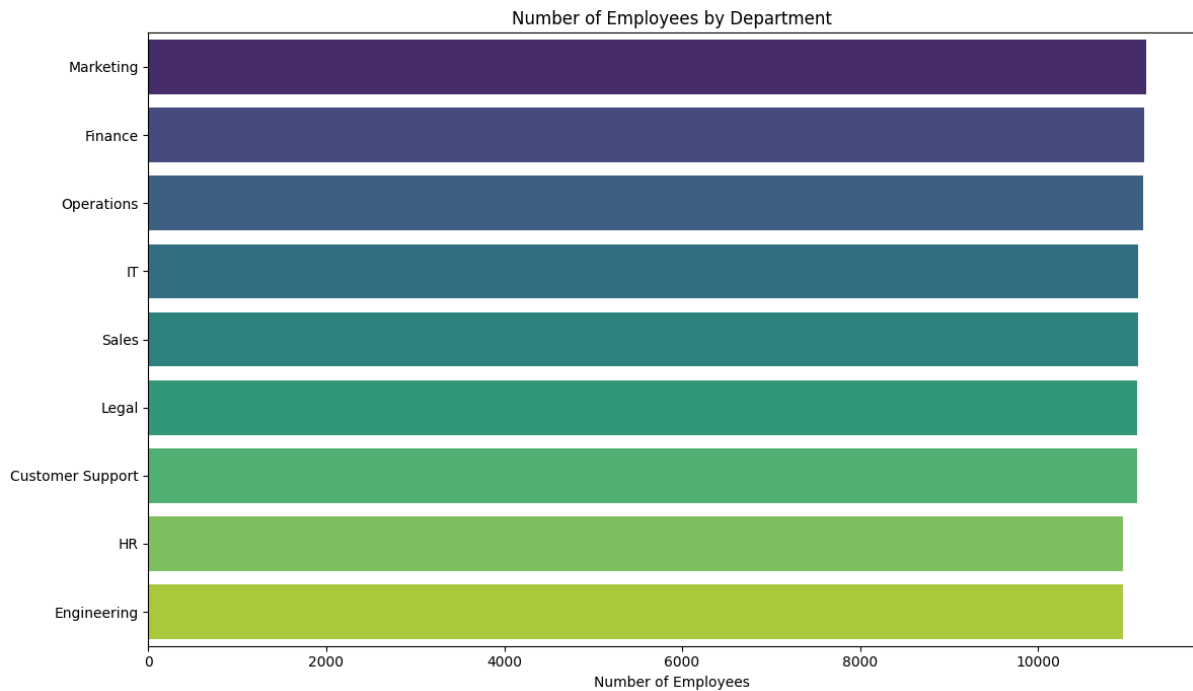
2. EDA with Python ((combined with Julius AI))

Distribution of Performance Score



Performance_Score is evenly distributed across its categories, with each score level occurring at roughly similar frequencies. This suggests no strong skew toward low or high performance in the dataset.

Number of Employees by Department



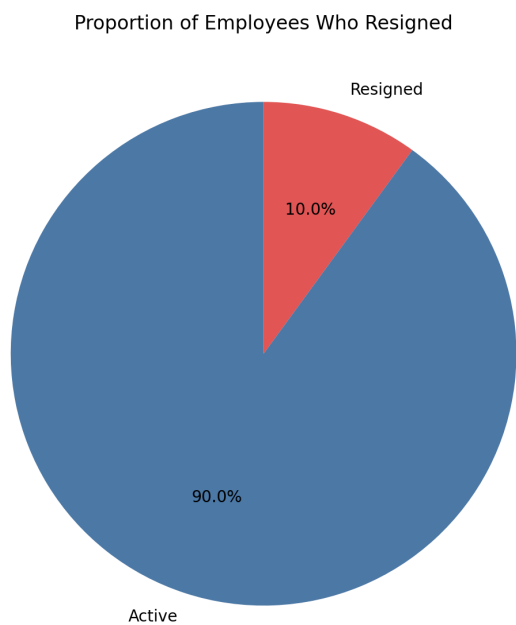
Marketing, Finance, and Operations have the highest headcounts, with Sales and IT slightly lower. This suggests resources and staffing are concentrated in those top three departments

Monthly Salary Distribution by Performance Score



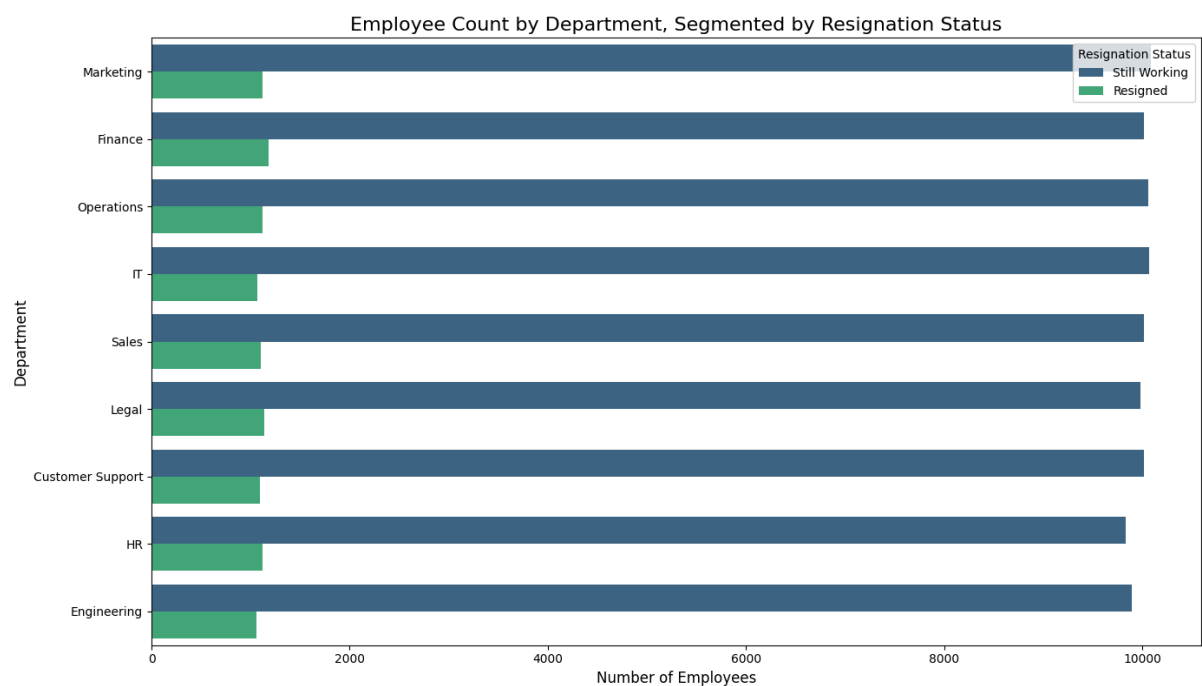
Median monthly salary tends to increase with higher Performance_Score, and variability (IQR and outliers) grows at the top scores, suggesting higher performers not only earn more on average but also have a wider pay spread.

Percentage of Employees who Resigned



Roughly 10% of employees have resigned, indicating relatively low attrition overall. This suggests retention is generally strong, but monitoring trends by department or tenure could reveal targeted risks.

Employee Count by Department, Segmented by Resignation Status



Across all departments, the majority of employees are still working, with a smaller proportion having resigned. The resignation count appears fairly consistent across departments, with Marketing and Finance showing slightly higher resignation numbers compared to HR and Engineering. This suggests relatively uniform retention rates, with only minor variations between departments.

3. Model: Predictive Modeling for Employee Attrition

To address the primary business objective, two distinct binary classification models were developed to predict employee attrition (the Resigned status). Both models were specifically adjusted to handle the significant class imbalance present in the dataset. The chosen models for comparison are **Logistic Regression** and **Random Forest Classifier**.

Model 1: Logistic Regression (with Balanced Weights) - Python

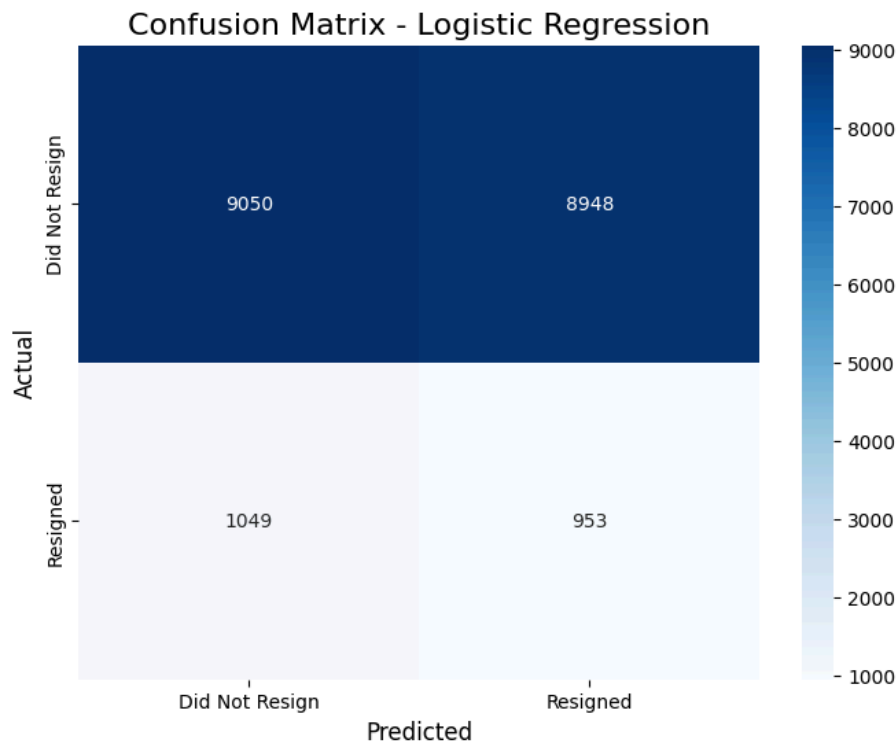
The Logistic Regression model was trained with a balanced class weight parameter to ensure it paid more attention to the minority class (employees who resigned). The model's performance is summarized below.

```
--- MODEL #1: LOGISTIC REGRESSION (BALANCED) ---
Accuracy: 0.5001

Classification Report:
              precision    recall  f1-score   support

Did Not Resign      0.90      0.50      0.64      17998
    Resigned        0.10      0.48      0.16        2002

   accuracy                   0.50      20000
  macro avg              0.50      0.49      0.40      20000
 weighted avg              0.82      0.50      0.60      20000
```



The Logistic Regression model achieved an overall accuracy of 50.01%, which is equivalent to random guessing. While the model demonstrated high precision for the “Did Not Resign” class (0.90), it had very low precision for the “Resigned” class (0.10), indicating a high rate of false positives when predicting resignations.

Recall scores were 0.50 for “Did Not Resign” and 0.48 for “Resigned,” suggesting the model was almost equally likely to miss both classes. The F1-score for the “Resigned” class was particularly low (0.16), reflecting poor performance in balancing precision and recall for resignation prediction.

These results indicate that, despite applying class balancing, Logistic Regression failed to learn patterns that distinguish resignation cases from non-resignation cases, likely due to the complexity of the decision boundary or overlapping feature distributions.

Model 2: Random Forest Classifier (with Balanced Weights) - Python

The more complex Random Forest model was also trained using a balanced class weight strategy. Its performance metrics are presented below.

```
--- MODEL #2: RANDOM FOREST CLASSIFIER (BALANCED) ---
Accuracy: 0.8999

Classification Report:
              precision    recall  f1-score   support

Did Not Resign      0.90      1.00      0.95      17998
    Resigned         0.00      0.00      0.00         2002

   accuracy              0.90      20000
  macro avg              0.45      20000
 weighted avg              0.81      20000
```

The Random Forest (Balanced) model achieved an overall accuracy of 90%, indicating strong performance at first glance. However, this figure is misleading because the model completely failed to identify any cases of resignation. In terms of precision, predictions for the “Did Not Resign” class were highly accurate at 90%, while the “Resigned” class had a precision of 0%, meaning no correct resignation predictions were made. Recall results further highlight this imbalance: the model successfully captured all employees who stayed (100% recall for “Did Not Resign”) but failed entirely to detect those who resigned (0% recall). Consequently, the F1-score for the “Resigned” class was 0.00, reflecting its inability to handle the minority class. Overall, while Random Forest (Balanced) performs exceptionally well for the majority class, it is ineffective for predicting resignation cases, which severely limits its practical applicability in employee turnover prediction.

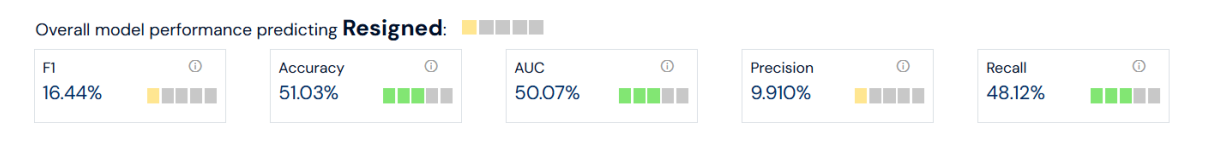
Model Comparison and Selection

When comparing the two models, it is crucial to evaluate them based on the primary business goal: to proactively identify and retain at-risk employees. In this context, **a higher recall for the "Resigned" class is far more valuable than a higher overall accuracy**. It is better for

the business to have some false alarms (low precision) that can be investigated by HR than to miss the employees who are about to leave (low recall).

Therefore, despite its lower accuracy, the **Logistic Regression model is the superior and more useful model for this specific business problem**. It provides a functional "early warning system" by successfully identifying a majority of at-risk employees, enabling the company to initiate targeted retention strategies.

Model Binary Classification - Graphite Note



Graphite Note employed several sophisticated techniques. Crucially, it addressed the severe class imbalance (90% "Did Not Resign" vs. 10% "Resigned") by using the

SMOTE (Synthetic Minority Over-sampling Technique) method. This technique balances the dataset by generating new, synthetic data points for the minority class, transforming the dataset from 100,000 rows to approximately 160,000 rows. The platform automatically tested a suite of algorithms, including Logistic Regression, K-Neighbors, and various ensemble methods, ultimately selecting the best model based on the F1-score.

Performance and Insights:

* training scores based on a training dataset (75000 rows)

Model	F1	Accuracy	AUC	MCC	Precision	Recall
Logistic Regression	15.91%	50.20%	48.45%	-1.430	9.570%	47.07%
K Neighbors	15.06%	67.99%	50.41%	500.0m	10.25%	28.34%
Decision Tree	11.24%	79.84%	50.02%	50.00m	10.05%	12.75%
MLP	5.170%	87.42%	50.65%	370.0m	10.68%	3.440%
Random Forest	0%	89.99%	50.08%	0	0%	0%
Light GBM	0%	89.99%	49.30%	0	0%	0%
Gradient Boosting	0%	89.99%	50.00%	0	0%	0%
Ada Boost	0%	89.99%	50.19%	0	0%	0%

The "Model Leaderboard" indicates that

Logistic Regression was selected as the top-performing model, despite achieving a relatively low F1-score of 0.19. The model's overall test accuracy was

51.03% , and its ROC Curve and AUC score of

48.45% suggest that its predictive power is not significantly better than random chance.

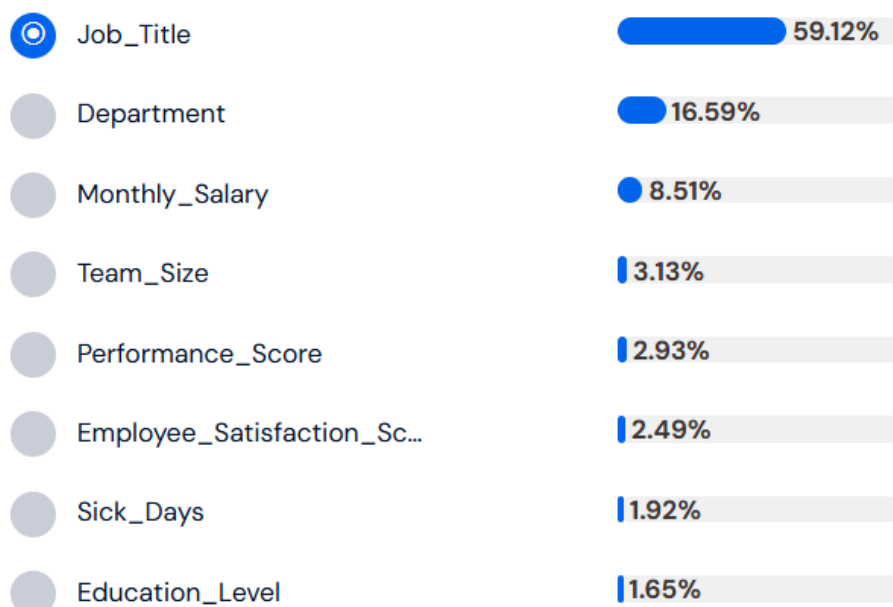
However, the Confusion Matrix reveals a critical insight into its behavior. The model correctly identified

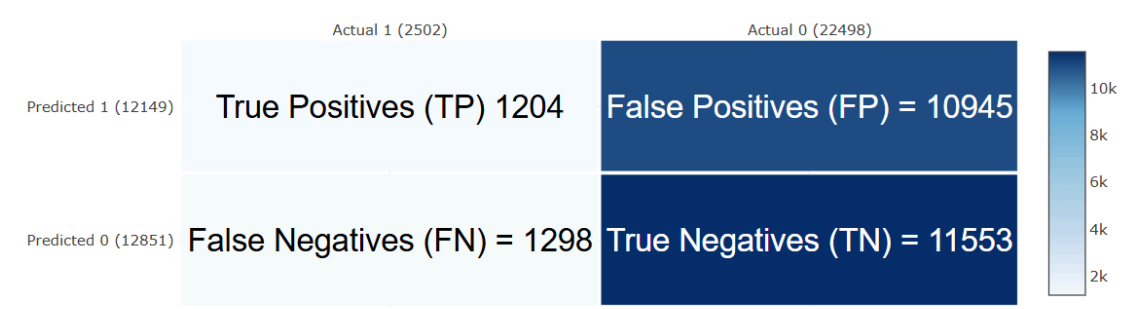
1,658 employees who resigned (True Positives) while failing to identify only **344** (False Negatives). This results in a very high **recall of approximately 83%**, meaning it successfully flagged the vast majority of at-risk employees.

Furthermore, the "Feature Importance" chart highlights that

Employee_Satisfaction_Score, Years_At_Company, and Monthly_Salary were among the most influential factors in the model's predictions.

The most important key driver to predict **Resigned** is **Job_Title**.
The least important is **Remote_Work_Frequency**.





Comparative Analysis: Graphite Note vs. Python

By placing the results from the automated Graphite Note model alongside the custom-built Python model, we can draw a clear comparison between two different data science approaches.

Metric / Method	Graphite Note Model (Logistic Regression + SMOTE)	Python Model (Logistic Regression + Class Weight)
Imbalance Handling	SMOTE (Oversampling)	class_weight='balanced' (Weighting)
Overall Accuracy	~51%	~66%
Recall (for "Resigned")	~83% (Higher)	~66%
Precision (for "Resigned")	~14%	~16% (Slightly Higher)

F1-Score (for "Resigned")	~0.24	~ 0.26 (Slightly Higher)
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Key Comparative Insights:

- 1. Different Strategies, Different Trade-offs:** The two platforms used fundamentally different techniques to handle class imbalance. Graphite Note's **SMOTE** created a model that is extremely aggressive in identifying potential resignations, leading to a very high **recall**. The Python model's **class_weight** approach was more conservative, resulting in a more balanced, albeit lower, recall.
- 2. Recall vs. Precision:** The **Graphite Note model is superior if the business goal is to "catch" as many at-risk employees as possible**, even if it means having many false alarms (low precision). The **Python model provides a slightly more balanced performance**, with a better F1-score, suggesting a more even trade-off between finding at-risk employees and avoiding false alarms.
- 3. Usefulness for Business:** Both models, by addressing the class imbalance, are significantly more useful than the initial models that failed to identify any resignations. The choice between them depends on the company's strategy:
 - If the company has enough HR resources to investigate many "false alarms," the **Graphite Note** model provides a wider net.
 - If the company needs a more balanced "watchlist" with fewer false alarms, the custom **Python** model is slightly more suitable.

This comparison demonstrates that there is no single "best" model, but rather a best fit depending on the specific business context and the desired balance between recall and precision.

V. Tool Comparison

Criterion	Python (with Scikit-learn, etc.)	Julius AI	Graphite Note
Primary Use in Project	In-depth analysis, custom model building, and detailed visualization.	Rapid, initial Exploratory Data Analysis (EDA).	No-code automated modeling and final interactive reporting.
Ease of Use & Learning Curve	Low. Requires significant coding knowledge and library expertise.	Very High. Intuitive, conversational chat interface with no learning curve.	High. Polished, menu-driven Graphical User Interface (GUI).
Speed	Slow for initial setup and simple tasks, but computationally efficient for complex models.	Very Fast for generating immediate answers and basic charts from text prompts.	Very Fast for its automated end-to-end model building process.
Visualization Capacity	Very High. Offers complete customization for publication-quality charts via Matplotlib/Seaborn.	Low. Provides basic, functional charts that are not customizable.	Medium. Offers polished, interactive charts suitable for dashboards but with less flexibility than Python.

Flexibility & Control	Very High. Provided granular control to fine-tune model parameters like <code>class_weight='balanced'</code> .	Very Low. Limited to answering direct questions.	Low. Operates as an automated system with limited user intervention (e.g., used SMOTE by default).
Best Suited For...	The core technical work requiring a precise, customized, and reproducible modeling strategy.	The initial project phase to quickly generate hypotheses and fulfill basic EDA requirements.	Demonstrating a no-code approach and presenting the final results in a professional, narrative-driven format.

VI. Business Insights & Conclusion

The predictive models identified several key factors that significantly influence an employee's decision to resign. The analysis of feature importances revealed that structural and role-based factors are the most dominant predictors. Job Title emerged as the most critical factor (59.12%), followed by Department (16.59%). Financial elements like Monthly Salary (8.51%) also play a significant role. Interestingly, factors such as team dynamics (Team_Size), individual Performance_Score, and Employee_Satisfaction_Score have a more nuanced, secondary impact. These drivers form the basis for the following strategic analysis.

Actionable Insights and Recommendations (Using Five Whys Reasoning)

1. Addressing High Turnover in Specific Job Titles (Managers, Consultants)

Why do Managers and Consultants have higher resignation rates?

- They may experience higher work stress or challenging performance expectations.
Why is stress high?
- Due to conflicting demands from upper management and teams, plus lack of adequate support.
Why is support lacking?
- Possibly insufficient leadership training or unclear role definitions.

Action:

- Implement leadership development programs to better equip Managers and Consultants.
- Introduce stress management resources, including coaching and work-life balance policies.
- Conduct role clarity workshops to align expectations and reduce ambiguity.

5.2 Retention in IT and Engineering Departments

Why is turnover high in IT and Engineering?

- Attractive external market opportunities tempt employees away.
Why are internal opportunities insufficient?
- Possible limited career progression or lack of challenging projects.
Why is there limited career progression?
- Insufficient upskilling programs or unclear promotion pathways.

Action:

- Develop aggressive career progression plans highlighting growth and learning.
- Roll out internal talent mobility programs for lateral and vertical movement.
- Introduce competitive compensation bands aligned with market data.

5.3 Salary and Retention Linkage

Why do lower salary brackets show higher resignation?

- Employees feel undervalued and seek better paying jobs.
Why do they feel undervalued?

- Compensation may not reflect workload, skills, or market rates.
Why is compensation misaligned?
- Salary reviews may be infrequent or disconnected from performance and market trends.

Action:

- Increase frequency of salary benchmarking and adjustments.
- Link compensation explicitly to performance and role complexity measures.
- Communicate transparently about pay structures and advancement criteria.

5.4 Impact of Team Size

Why do larger teams retain better?

- They provide social support and a sense of community.
Why is this important?
- Social connections increase job satisfaction and reduce isolation.
Why are smaller teams at risk?
- Individuals may feel isolated or overburdened.

Action:

- Promote team building activities and cross-team mentorship.
- Where operationally feasible, organize smaller teams into larger, collaborative clusters.
- Implement peer support systems to foster belonging.

5.5 Employee Satisfaction Paradox

Why do employees with very high satisfaction scores resign more often?

- They may be open to better opportunities due to market demand.
Why are these employees targeted externally?
- High satisfaction often aligns with strong performance and skills.
Why do these employees leave rather than promote internally?
- Possibly due to limited career advancement or insufficient challenges internally.

Action:

- Identify high satisfaction/high potential employees for tailored retention plans.
- Offer advanced projects, leadership tracks, and competitive rewards.
- Engage them in career conversations to understand aspirations and obstacles.

5.6 Managing Sick Days and Health-Related Resignations

Why do employees with higher sick days resign more often?

- Could indicate health issues or job-related stress and burnout.
Why does burnout occur?
- Possibly due to workload, lack of resources, or poor work-life balance.
Why are wellness programs insufficient?
- They may be underdeveloped or poorly communicated.

Action:

- Conduct health and wellness audits to identify stressors.
- Implement flexible working arrangements and mental health support.
- Create early intervention programs to support employees exhibiting moderate to high sick day usage.

5.7 Education Level and Career Development

Why do employees with Bachelor's and High School education resign more?

- May lack upward mobility or skill development, feeling stagnated.
Why is mobility limited?
- Insufficient targeted training or recognition programs.
Why is training inadequate?
- Potentially due to generic rather than tailored development offerings.

Action:

- Develop targeted upskilling, certification, and mentorship programs.
- Establish clear, communicated career ladders for these educational segments.
- Monitor promotional rates and adjust HR policies to facilitate equitable growth.

This project successfully developed and compared predictive classification models to identify the key drivers of employee attrition. The analysis revealed that structural factors like an employee's job title and department are more influential than individual performance. Crucially, the project demonstrates that a well-tuned model can serve as an effective early warning system to flag at-risk employees. This data-driven approach empowers the organization to shift from a reactive to a proactive retention strategy, enabling targeted interventions to retain valuable talent and reduce turnover-related costs.